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Contextual feature extraction hierarchies converge in large language models and the brain

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Recent advancements in artificial intelligence have sparked interest in the parallels between large language models (LLMs) and human neural processing, particularly in language comprehension. Although previous research has demonstrated similarities between LLM representations and neural responses, the computational principles driving this convergence—especially as LLMs evolve—remain elusive. Here we used intracranial electroencephalography recordings from neurosurgical patients listening to speech to investigate the alignment between high-performance LLMs and the language-processing mechanisms of the brain. We examined a diverse selection of LLMs with similar parameter sizes and found that as their performance on benchmark tasks improves, they not only become more brain-like, reflected in better neural response predictions from model embeddings, but they also align more closely with the hierarchical feature extraction pathways of the brain, using fewer layers for the same encoding. Additionally, we identified commonalities in the hierarchical processing mechanisms of high-performing LLMs, revealing their convergence towards similar language-processing strategies. Finally, we demonstrate the critical role of contextual information in both LLM performance and brain alignment. These findings reveal converging aspects of language processing in the brain and LLMs, offering new directions for developing models that better align with human cognitive processing.

The intersection of artificial intelligence and neuroscience has emerged as a frontier of great interest, particularly for understanding how large language models (LLMs) and the human brain process language. Prior research has laid the foundation in this area, uncovering intriguing parallels in feature extraction and representational similarities between LLMs and neural responses during language processing. Studies[1–8] have demonstrated that representations learned by LLMs can be linearly mapped to neural responses, suggesting that both LLMs and the brain utilize comparable features in language processing. However, these findings offer limited insights into the fundamental characteristics of LLMs that enable this brain-like processing.

Further investigations have delved into the different aspects of LLMs to elucidate their resemblance to brain processes. Some studies[9,10] have supported the predictive coding hypothesis in human language processing by finding stronger similarities with autoregressive LLMs. Others[4,7,11,12] have explored various factors, such as LLM language modelling performance, model size and capacity and the generalizability of linguistic representations as indicators of brain-like processing. These studies imply that the quality of an LLM substantially contributes to its brain-like representations. However, the underlying reason for this similarity remains an open question. Is it merely a matter of scaling up the models[11], or do these models share fundamental

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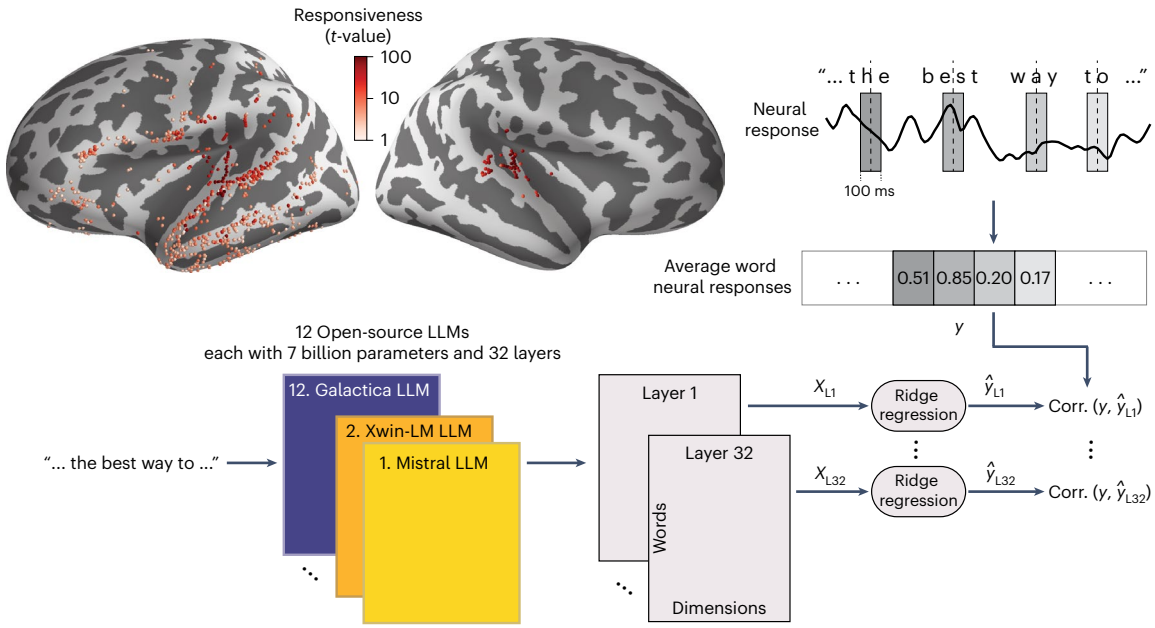

Fig. 1 | Mapping LLM embeddings to the brain. Speech-responsive electrodes are shown on an inflated brain (shaded by their responsiveness *t*-value from a paired *t*-test between speech and silence). As participants listened to speech, the average neural response in a 100-ms window around a word centre was used as the word response of a given electrode. The same text was fed to an LLM, and the embeddings from all 32 layers were extracted. Ridge regression was used to predict the word responses from the LLM representations, producing a brain correlation (corr.) score for each electrode–layer pair.

computational principles that increasingly align well with the spoken language-processing pathway in the human brain? This question is critical because it suggests a potential shift in the paradigm of model optimization. Although both brains[13–16] and LLMs[17,18] process speech and language in hierarchical pathways, most studies have analysed the similarity of their representations without detailed comparisons of the hierarchical processes through which they are created. Thus, it remains unclear whether brains and models arrive at similar representations through the same or different pathways.

We aimed to answer these questions by examining the interplay between LLM performance, neural predictability, anatomical alignment and contextual encoding, potentially paving the way towards models that perform with high accuracy and process language in a manner similar to the human brain. We examined 12 open-source, pretrained LLMs, all uniform in size but diverse in their linguistic capabilities, particularly in language understanding tasks, such as reading comprehension. We recorded neural responses using intracranial electroencephalography (iEEG) in the auditory cortex and speech-processing regions of neurosurgical patients as they listened to speech. We then predicted these neural responses from the embeddings extracted from each layer of the LLMs as they processed the same linguistic input. This approach allowed us to pinpoint which layers and aspects of the LLMs were most predictive of brain activity and to explore how variations in model performance align with differences in neural prediction and their anatomical and functional correspondence. Our findings offer a fresh perspective on the evolving landscape of LLMs, providing insights that reveal intricate parallels in language comprehension between artificial and biological systems, uncover potential reasons for LLM performance differences and point to a convergence in LLMs towards a more optimal, brain-like language-processing system.

Results

Brain similarity of LLMs

We studied 12 recent, popular, open-source LLMs, all with approximately seven billion parameters and 32 layers. We evaluated each model on a suite of benchmark tasks to assess its language modelling performance, splitting these tasks into categories relevant to English language comprehension, specifically reading comprehension and common-sense reasoning, as in a previous study[19] (see Methods and Table 1 for exact benchmarks). Overall LLM performance was estimated as the average score for these two categories.

Neural responses were recorded using invasive electrodes (iEEG) from eight neurosurgical patients, with electrode placement determined by clinical need (Supplementary Fig. 1). The participants listened to 20–30 min of speech from various speakers, including stories voiced by voice actors and dialogues between characters. The text for each audio file was fed into each LLM, and we extracted the causal embeddings of each word in every layer. We reduced these embeddings to 500 components using principal component analysis (PCA) to ensure consistent dimensionality. We used ridge regression to estimate the similarity of the features of the model to the brain (Fig. 1)[3,7,9]. We analysed 707 electrodes that were responsive to speech, as determined by a two-sided *t*-test between responses to words and silence (false discovery rate corrected; $P < 0.05$ (ref. 20)). For each responsive electrode, we extracted the average high-gamma band envelope response in a 100-ms window around the centre of every word. Sampling around word centres ignores dynamics, such as hierarchical temporal integration and receptive fields[14,15,21,22], which likely influence the ideal neural response period of each electrode. However, word centres approximately match the population of electrode response lags (Supplementary Fig. 2), allowing a standardized neural response sampling point across electrodes without introducing additional variability from potentially noisy lag estimation. We then fitted cross-validated ridge regression models to predict these neural responses from the word embeddings and used the average prediction correlation on the withheld folds as the brain similarity with that electrode. Neither the number of principal components of the embeddings nor the window size used to compute the neural response to words affected the results (Supplementary Fig. 3).

The electrode-averaged brain similarity across the layers of each model is shown in Fig. 2a. With these latest LLMs, we confirmed previous findings showing that neural responses can be predicted from model representations, and we found that brain similarity

Table 1 | All models used in this study, along with their computed benchmark performance

Models used	Reading Comprehension	Commonsense Reasoning	LLM performance
Galactica-6.7B[82]	0.486	0.535	0.511
CerebrasGPT-6.7B[83]	0.565	0.575	0.570
Pythia-6.9B[84]	0.568	0.597	0.582
OPT-6.7B[85]	0.581	0.616	0.598
FairseqDense-6.7B[86]	0.575	0.628	0.602
LeoLM-7B[87]	0.634	0.646	0.640
MPT-7B[88]	0.620	0.665	0.643
Falcon-7B[89]	0.619	0.669	0.644
LLaMA-7B[90]	0.626	0.674	0.650
LLaMA2-7B[19]	0.639	0.671	0.655
Xwin-LM-7B[91]	0.648	0.673	0.660
Mistral-7B[92]	0.669	0.703	0.686

generally increases over layers and peaks in middle or later layers[3,7]. Higher-performance LLMs also achieved higher peak brain scores (Pearson's $r = 0.92$; $P = 2.24 \times 10^{-5}$) (Fig. 2b), indicating that they extract more brain-like features from language.

Similar to the layers of a model, the auditory and language-processing pathway demonstrates hierarchical organization[13–15,23]. The primary auditory cortex, the first point of auditory processing in the cortex, is centred around the posteromedial Heschl's gyrus (pmHG or TE1.1)[24]. Because this is a common reference point in auditory cortical processing, we quantified the depth of each electrode in the spoken language-processing pathway of the brain using its geodesic distance from this landmark along the cortical surface[25–27] (Supplementary Fig. 4). Previous studies have found that deeper layers of LLMs correspond better to deeper language-processing regions of the brain[7,10,28]. We confirmed this result (Fig. 2c), but interestingly, we also found that better-performing LLMs peak in brain similarity at earlier layers compared to worse models (Pearson's $r = -0.81$; $P = 0.0013$) (Fig. 2d). This uncovered a new dimension in the evolution of LLMs: the progression of feature extraction over layers aligns differently with the brain for higher-performance versus lower-performance models.

Alignment of model and brain linguistic hierarchies

Given that the layer-wise brain similarity of the models with higher LLM performance scores appears different from that of the models with lower LLM performance scores, we hypothesized that higher-performance models were not only learning more brain-like features, but that the progression of feature extraction performed by these models was different. Inspired by an investigation of hierarchical correspondence between stages of visual cortex processing and image classification networks[29], we sought to compute the alignment between hierarchical feature extraction pathways in the brain and models. Although the exact hierarchical processing stages of the brain, analogous to the layers of a model, are not perfectly known, we again used the distance from pmHG to quantify the stages of hierarchical processing. We grouped the electrodes into bins at 10-mm intervals. We normalized the brain similarity scores of each electrode over the layers before averaging these scores over the electrodes in a bin to produce a set of layer scores, which are shown as a single row of the alignment matrix in Fig. 3a. We used the centre of mass of this average brain similarity score over layers within each electrode bin to quantify the depth in the LLM corresponding to this given stage of the hierarchy of the brain (see Methods for further details). We then compared the progression of these most similar LLM layers to the bin distances along the hierarchy, visually finding that some models achieved a more linear increase in LLM layers over the bins. We summarized the alignment between the language-processing hierarchies of each LLM and the brain using the Pearson's correlation between the layer centre of mass in each bin and the hierarchical stage of each bin (that is, the distance of each bin from pmHG)[29]. We illustrate this alignment computation for Xwin-LM and Galactica, the two models that achieved the highest and lowest hierarchy alignment scores, respectively (Fig. 3a). These models also displayed a stark difference in benchmark performance, with Galactica being the lowest-performance LLM. The alignment scores reveal that the better model (Xwin-LM) exhibits a feature extraction progression that is more consistent with the brain from early- to late-stage processing than the worse model. This brain alignment is also highly correlated with LLM performance on benchmark evaluation tasks (Pearson's $r = 0.79$; $P = 0.0021$) (Fig. 3b). We found the same result when using electrode latency to measure the stages of the hierarchical processing of the brain, rather than the distance from the pmHG (Pearson's $r = 0.89$; $P = 0.0001$) (Supplementary Fig. 5), which demonstrates that this finding holds for other estimates of the stages of the cortical hierarchy. Additionally, to ensure that this effect was not the result of a single participant overpowering the distribution, we separated the even- and odd-numbered participants and performed the analysis again, finding that the brain hierarchy alignment was significantly correlated with LLM performance for each group (Pearson's correlation, even participants: $r = 0.79$ and $P = 0.0022$; odd participants: $r = 0.81$ and $P = 0.0013$) (Supplementary Fig. 6). Overall, these findings demonstrate that better-performing LLMs extract features using a hierarchy that aligns more linearly with the hierarchical language-processing pathway of the brain.

To perform model-to-model comparisons, we used centred kernel alignment (CKA)[30], a method analogous to canonical correlation analysis (CCA) with a nonlinear kernel, which can capture the similarity between high-dimensional representations, such as neural network embeddings. We computed the CKA similarity between all pairs of layers for all pairs of models. Thus, each pair of models creates a layer-by-layer similarity matrix that describes their embedding similarities. The high similarity along the diagonal indicates that the two models extracted similar features in the same layers. A higher similarity offset from the diagonal indicates that one model exhibits a delay in extracting similar features. When grouping these similarity matrices by the top 5 and bottom 5 models based on the LLM benchmark performance and averaging within a group, an interesting pattern emerged (Fig. 4a). We found that the top models exhibited a high degree of similarity to each other along the diagonal. On the other hand, the worst models were much less similar to each other in their early layers, and even in their later layers, they were less consistent than the top-5-to-top-5 model pairs. Finally, comparing the top 5 models to the bottom 5 models revealed a striking offset in the maximum similarity from the diagonal. This suggests that bad models require more layers to reach a similar level of feature extraction as good models. We summarized the layer-wise feature extraction similarity between each pair of models using the average CKA similarity along the diagonal in the CKA similarity matrix (Fig. 4b). The plot demonstrates that the top 5 models were indeed more similar to each other, with a sub-block of high similarity emerging among the top few models. Because Mistral is the best-performing LLM, we looked at the diagonal similarity to Mistral of each model and found that a more Mistral-like feature extraction progression correlated strongly with LLM performance (Pearson's $r = 0.79$; $P = 0.0022$) (Fig. 4c). These results revealed distinctions between the embeddings of LLMs and suggest that inefficient feature extraction or poor early-layer learning in bad models may contribute to their worse performance and lower brain similarity.

Contextual content supports brain hierarchy alignment

Because the contextual nature of LLM features is critical for their brain similarity compared with non-contextual representations[3,9,31],

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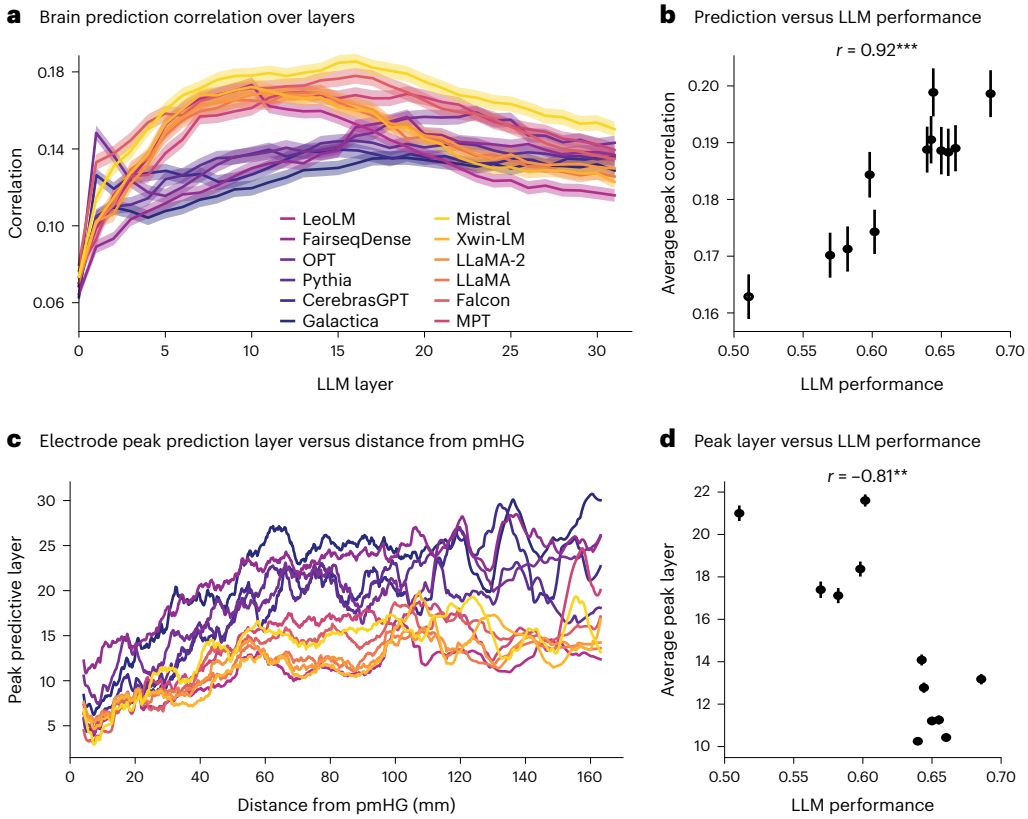

Fig. 2 | Peak brain correlations and layers related to LLM performance. **a**, Average brain correlation over all electrodes for each LLM. LLMs are coloured in the order of their separately measured benchmark performance, with blue/purple models performing the worst and yellow models performing the best. The shaded regions indicate the standard error of the mean over the electrodes. **b**, Peak correlation over all layers of a given model was computed for each electrode and then averaged over all electrodes. The bars indicate the standard error of the mean over the electrodes. The average peak correlation score was significantly related to LLM performance (Pearson's $r = 0.92$; $P = 2.24 \times 10^{-5}$; $n = 12$). $^{**}P < 0.01$ and $^{***}P < 0.001$. **c**, The peak scoring layer of each model was computed for each electrode. Electrodes were then sorted by distance from the pmHG, and a sliding window average (centred; $n = 50$) was taken across the electrodes of each model to compute the smoothed local estimate of the most brain-like LLM layer. The peak scoring layer generally increases with distance from the pmHG, and the better models (yellow) peak at lower layers compared to the worse models (blue/purple). **d**, The average peak layer in terms of neural prediction performance for a given model over all electrodes is shown with bars indicating the standard error of the mean. The average peak layer was significantly negatively related to LLM performance (Pearson's $r = -0.81$; $P = 0.0013$; $n = 12$).

we hypothesized that the amount of contextual information used by a model may also play a key role in determining the alignment between the hierarchical feature extraction pathways of LLMs and the brain. We extracted limited-context embeddings from the LLMs by restricting their causal attention mechanism to a certain window in the previous text. Transformer architecture LLMs use tokenizers to separate text into discrete units; therefore, we supplied the models with only the most recent N tokens, sweeping N over a range of values from 1 to 100. A single-token input gives the model no context at all. For reference, in our corpus, the Mistral tokenizer splits one word into 1.15 tokens on average. We then repeated our analysis of the model–brain hierarchical alignment (as previously shown in Fig. 3b) by computing the correlation between brain hierarchy alignment and LLM performance at each limited context window length. Although this correlation is positive for all cases except the one-token case, it is only significant for long contextual window lengths of 50 tokens and above (Fig. 5a). This suggests that the brain alignment of LLMs critically depends on the amount of contextual information that the model can see, which influences its hierarchical feature extraction mechanism.

Because the correlation between LLM performance and hierarchical alignment is strongly positive for long context lengths, we expected that better-performing models would be better at incorporating contextual information into their language representations. To test this, we quantified the amount of contextual information present in the embeddings of a model by measuring how much the embeddings changed when contextual information was added to the input. We measured the CKA difference (1 − $\text{similarity}_{\text{CKA}}$ (full context; one token)) of the embeddings of each layer when given the full context, compared to the first-layer embeddings when given only a one-token limited context window. We refer to the average of this CKA difference across all layers as the contextual content of the representations of the model. We found that this contextual content was positively correlated with LLM performance (Spearman's $r = 0.66$; $P = 0.020$) (Fig. 5b). Additionally, it was very strongly correlated with brain similarity (Spearman's $r = 0.84$; $P = 0.0006$) (Fig. 5c). These findings indicate that contextual information plays a crucial role in natural language processing in both natural and artificial language models, and that contextual feature extraction enables brain hierarchy alignment in LLMs.

We further investigated the impact of contextual information on neural similarity by computing the extent to which each LLM's peak similarity score with a given electrode changed when the models were given full context versus no context (one token). We then averaged this difference over all LLMs for each electrode and plotted the electrodes on the brain, finding that being given extra context greatly improved similarity scores with electrodes in higher-level language-processing areas (Fig. 5d). Averaging electrodes within major anatomical regions further quantified this result because we found higher average context effects on brain correlation score within the

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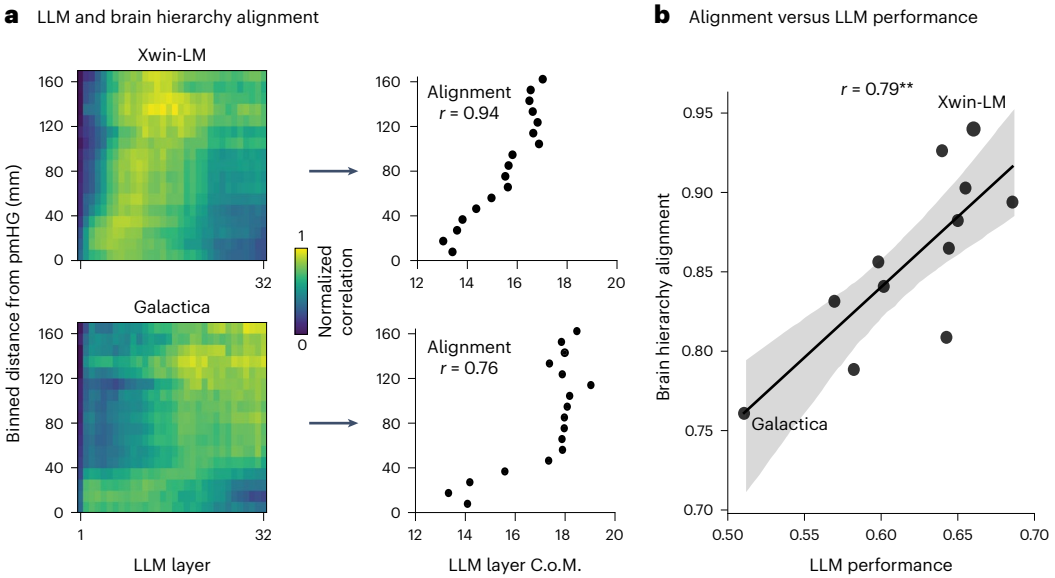

Fig. 3 | Better LLMs display more brain-like hierarchical processing. **a**, Examples of computing the brain hierarchy alignment are shown for two models: Xwin-LM (the model with the highest alignment score) and Galactica (the model with the lowest alignment score). The electrodes were first binned into a hierarchy based on their distance from the pmHG. Within a bin, the correlations over all 32 layers were normalized between 0 and 1 and then averaged over electrodes in the bin, producing one row for each bin in the matrix on the left. The centre of mass (C.o.M.) of the distribution of brain similarity scores over the LLM layers for each bin was computed and plotted in a scatter plot to the right. The brain hierarchy alignment score was then computed as the Pearson's correlation between the LLM layer centre of mass and the distance from the pmHG. **b**, A scatter plot of brain hierarchy alignment scores and LLM performance shows a significant positive correlation (Pearson's $r = 0.79$; $P = 0.0021$; $n = 12$; ** indicates $P < 0.01$). The line shows a linear regression fit, and the shaded region indicates a bootstrapped ($n = 1{,}000$) 95% confidence interval.

higher-level linguistic-processing area of the inferior frontal gyrus (IFG)[32] compared to sensory regions, such as Heschl's gyrus (HG) and superior temporal gyrus (STG) (Wilcoxon rank-sum test; $P = 0.0079$ (HG) and $P = 6.04 \times 10^{-5}$ (STG); $N = 48$ (HG), 151 (STG) and 62 (IFG)) (Fig. 5e). Interestingly, the articulatory region of the subcentral gyrus ($N = 16$ electrodes), which has also been implicated in high-level linguistic processing[33], displayed the highest average score improvement, but due to its high variance, it did not meet statistical significance compared to the other regions. These results show that contextual information becomes more critical in determining brain similarity further along the spoken language-processing hierarchy, which supports previous investigations of high-level linguistic feature encoding in more downstream regions[34,35]. This finding strengthens the notion that both the brain and LLMs extract context along their hierarchies, and that LLMs need contextual information to achieve brain similarity in downstream processing regions. Taken together, our analyses revealed that high-performance LLMs not only extract representations of language that are similar to the brain, but they also use hierarchical feature extraction pathways that more strongly align with that of the brain due to contextual information-processing abilities, a finding that uncovered ways in which the best LLMs continuously converge towards the brain.

Discussion

We explored LLMs and their alignment with neural responses during language processing and uncovered several key findings. First, we observed a clear correlation between the language task performance of LLMs and their accuracy in predicting neural responses in the auditory cortex, with higher-performance models exhibiting greater functional alignment with the speech cortex. Second, we showed that models with higher performance on benchmark tasks achieved peak predictive accuracy in earlier layers. In contrast, lower-performance models exhibited a delayed representation, necessitating deeper layers to approach similar levels of brain prediction accuracy. Finally, our study highlights the crucial role of contextual information in both LLMs and brain processing, where the size of the contextual window strongly influences the difference between better and worse models, with the availability of long-range contextual information driving high-performance LLMs closer to the hierarchical pathway of the brain. These findings uncovered fundamental principles in language processing, highlighting the critical role of hierarchical structure and contextual dependencies in language that give rise to convergent processing strategies in both artificial and biological systems.

Hierarchical processing and inter-model comparisons

We found that better-performing LLMs exhibit a more brain-like hierarchy of layers, offering insights into their language processing. Although previous studies have revealed similarities in the hierarchical stages found in the brain and deep neural networks for linguistic[7,10,28], acoustic[36,37], visual[38–40] and imagined stimuli[41], a distinct approach in our study is the inter-model comparison within a consistent architectural framework. In a related study analysing deep neural networks for vision tasks, recent evidence[29] has shown that better performance can create a less brain-like progression of feature extraction in models compared to the visual cortex, suggesting that the complex architectures of high-performance image-processing networks have steered them away from neural alignment. By examining LLMs based on a single architecture, the stacked transformer decoder[42], we uncovered differences in their alignment with the hierarchical stages of the brain during language comprehension. Transformer language models use contextual features to encode linguistic, syntactic and positional structures[43,44], and increasingly high-level and context-specific features arise throughout the layers of the model[17,18]. This may be partly because later layers bind linguistic structures over longer contexts[45]. The crucial observation that such models display brain-like hierarchies resonates with neurobiological findings of hierarchical organization in the auditory and language-related cortex[13–16,23,34,46,47]. The convergence of the two systems highlights the inherent hierarchical structure of language as

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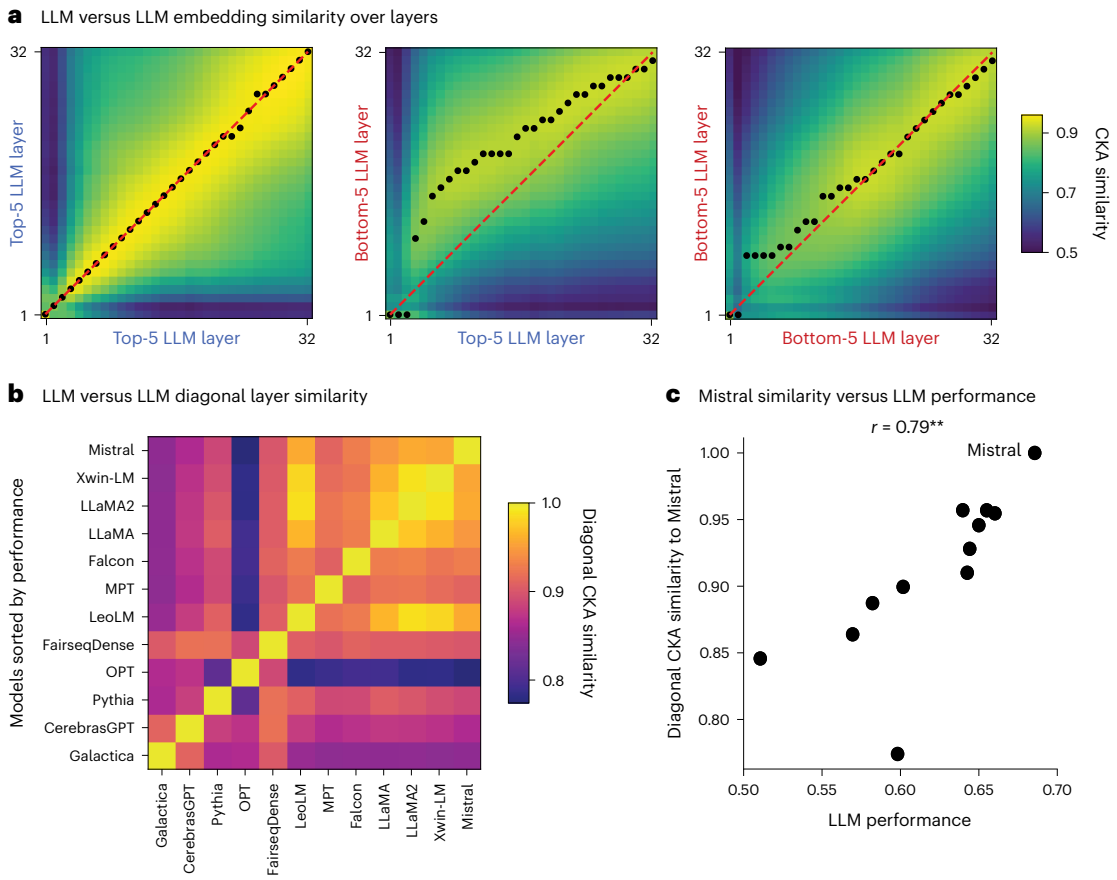

Fig. 4 | Comparison of feature extraction hierarchies between LLMs. **a**, Layer-by-layer similarity matrices were computed using CKA for every pair of LLMs. LLMs were labelled as either top 5, bottom 5 or excluded, depending on their sorted performance on our LLM benchmark evaluation. Similarity matrices between all pairs of the top 5 LLMs were then averaged and displayed as the 'top 5 versus top 5' average similarity matrix in the top left. The same was done to create similarity matrices between the average 'top 5 versus bottom 5' as well as the average 'bottom 5 versus bottom 5' LLMs. The layer of maximum similarity in every layer column is indicated by a black dot. Visually, from the locations of the peak similarity dots and the magnitude of the similarity, the 'top 5 versus top 5' similarity matrix is highly diagonal, whereas the 'bottom 5 versus bottom 5' is less similar in the early layers. The 'top 5 versus bottom 5' similarity matrix shows an offset diagonal, indicating a delay in feature extraction for the bottom 5 models compared to the top 5. **b**, Model-by-model diagonal similarity was computed as the average along the diagonal of the similarity matrix. The models are arranged in the sorted order of the LLM benchmark performance from worst to best. This visually confirms that the best models are fairly similar to each other in layer-wise feature extraction, whereas the worst models are less similar to each other and less similar to the best models. **c**, The diagonal similarity of each model with Mistral, the best-performing LLM, is plotted against LLM performance, showing a strong positive relationship (Pearson's $r = 0.79$; $P = 0.0022$; $n = 12$; $^{**}P < 0.01$).

we increasingly form larger units of representation, from articulatory features to phonemes, syllables, words, sentences and phrases[35,48,49]. In some ways, the increasing alignment between the models and the brain may be surprising because our characterization of the hierarchy of the brain using geodesic distance is a rough estimate that cannot account for complex white matter pathways between distant language regions, such as the arcuate fasciculus[50,51]. However, geodesic distance provides a good estimate of hierarchical connectivity without introducing extra parameters or assumptions[52]. Regardless, our results demonstrate that as LLMs have achieved higher performance, they have done so using feature extraction pathways that more closely resemble the human brain.

Feature extraction efficiency and contextual processing

A notable finding of our study is the delay in feature extraction across layers, which was observed in less effective LLMs compared to their higher-performance counterparts. This delay, particularly evident in the early processing stages of transformer models, suggests a slower buildup of relevant linguistic and contextual information[18]. The implications of this observation are multifaceted. First, it challenges the conventional emphasis on the final layers of LLMs[9], instead drawing attention to the critical role of the initial layers in efficient language processing[12]. This shift in focus aligns with emerging neuroscience research that underscores the significance of early-stage processing in the human brain for complex cognitive tasks, such as language processing[35,47,49]. Second, this delayed representation in less effective models offers insights into potential inefficiencies in their training or design. Given the architectural similarity of the models in our study, the variance in feature extraction efficiency among models may reflect differences in training strategies[53] and data quality[19,54,55], providing insights for future LLM model development. As LLMs have evolved in recent years, improvements in dataset size and cleanliness, as well as architectural changes to increase context length, have come along with their performance improvements, and our results show that these improvements have also given rise to greater brain similarity. Furthermore, the observation that higher-performance models utilize early layers more effectively and peak in their brain similarity in middle layers rather than in later layers raises intriguing questions about the role of subsequent layers. It is possible that these later layers are engaged in next-level contextual integration and feature extraction, which is potentially analogous to higher-order stimulus integration to support cognitive functions in the human brain[56,57]. Alternatively, this finding could point to a limitation in our current methodologies, such as limited

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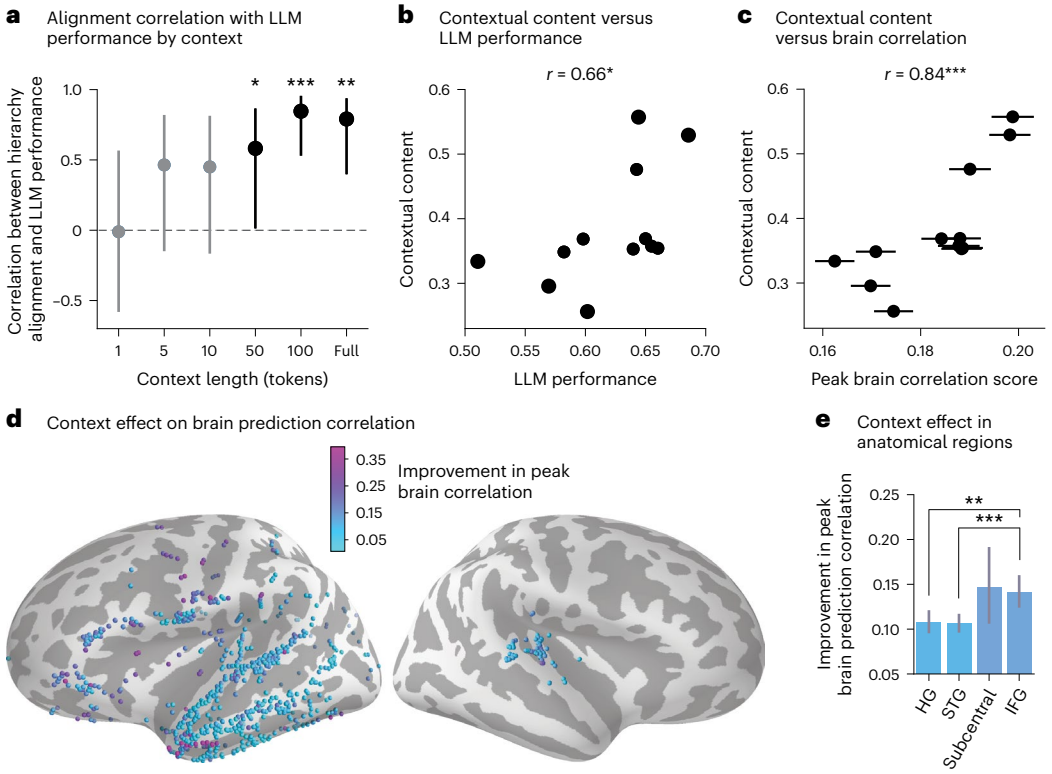

Fig. 5 | Effect of contextual information. a, Using embeddings from LLMs when given a limited number of previous tokens as a context window, we analysed brain hierarchy alignment again. Each dot represents the correlation between LLM performance and brain hierarchy alignment with 95% confidence interval bars (Pearson's correlation; $n = 12$), showing only a significant correlation for long contextual windows. $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$. **b**, The contextual content of the representations of a model is plotted against its benchmark performance, showing a positive correlation between the two (Spearman's $r = 0.66$; $P = 0.020$; $n = 12$). **c**, The contextual content of each model is plotted against its average peak brain similarity over the electrodes, showing a strong correlation (Spearman's $r = 0.84$; $P = 0.0006$; $n = 12$). Horizontal lines show the standard error of the mean over the electrodes for brain similarity. **d**, Electrodes plotted on the FreeSurfer average inflated brain, coloured by the effect of contextual information on the peak brain similarity score. **e**, Bar plot of the average context effect on the peak brain similarity score for electrodes within four main anatomical regions along the linguistic hierarchy. Each bar is coloured by its value according to the same colour map used in the brain plot, and the error bars show the standard error of the mean. Stars indicate significant differences between a pair of regions (Wilcoxon rank-sum test; $P = 0.0079$ for HG–IFG; $P = 6.0 \times 10^{-5}$; $n = 48$ (HG), 151 (STG), 16 (subcentral) and 62 (IFG)), $^{**}P < 0.01$ and $^{***}P < 0.001$.

iEEG coverage, the simplicity of the speech comprehension task or the fact that LLMs are not explicitly trained to perform comprehension, but rather next-word prediction, which is slightly different from the speech listening comprehension task performed by the participants. Our iEEG recordings include broad coverage of speech-processing regions, especially acoustic sensory regions, such as HG and STG, which, although critical for spoken language processing, represent a slightly different aspect of linguistic feature extraction than the token-level processing that transformer architecture LLMs begin with. Answering these questions is crucial to enriching our understanding of artificial language processing.

The influence of contextual information on brain similarity and LLM benchmark scores also points to specific avenues that may improve model performance in language tasks. Ensuring that models are able to extract long contexts, such as by using architectures that allow for long context windows[58] and utilizing training data rich in long context information, could enhance LLM performance further beyond simply scaling up the parameter size of a model. Transformer-based LLMs have been shown to suffer from unequal contextual information extraction when the prior context occurs at different distances from the target[59], supporting the notion that improving the robustness of modern LLMs to varying context lengths may lead to performance improvements. Our investigation offers a unique lens through which to view the parallels and divergences between machine learning and human cognitive development.

This raises a key question. What differentiates high-performance LLMs and enables them to perform better and in more brain-like ways in the first place? Unfortunately, we are limited in our ability to explain these differences because most models do not have extensive descriptions of their training data or how they were trained. Thus, we hypothesize that the major differences are caused by data quality[19,54,55], where better models were trained on data that were more thoroughly cleaned, such as removing duplicates or bad segments from text scraped from the internet, and potentially by optimization tricks, such as pretraining. Uncovering the differentiating factors between models is an important avenue for future research that will strengthen our understanding of LLM–brain similarity.

LLM convergence to brain-like processing

The convergence of LLMs and human speech processing may suggest that certain fundamental principles underlying efficient language processing might be common to both artificial and biological systems. The language capabilities of the human brain have developed as an adaptive response to complex communication needs, optimizing for efficiency and versatility[60]. Our findings suggest that LLM architectures and processing strategies are gravitating towards these same principles, mimicking the evolutionary adaptations of the brain for language. LLMs are trained without consideration for brain similarity; however, they have become increasingly brain-like in their feature extraction and hierarchical processing. Brain-like processing may represent an

optimal solution to language modelling found by evolution[61], although subject to biological constraints. Our results suggest that modern LLM training focused on performance optimization may have placed these models on a similar path. In our study, Mistral, the top-performing model, stands as a prime example of this convergence, in which the degree of similarity of the embeddings of a model to those of Mistral is highly correlated with performance and brain similarity. On the other hand, our naturalistic stimuli, although indicative of real-world listening environments, may have rarely presented infrequent linguistic structures or word choices that could have further differentiated the LLMs[62], suggesting a direction for future studies.

Our study marks an important step in understanding the parallels between LLMs and human brain processes in language comprehension by revealing the intricate relationship between internal model representation, model performance and neural predictive accuracy. Our findings enhance our understanding of LLMs and offer insights into the cognitive mechanisms underlying human language processing.

Methods

Human intracranial recordings

Eight individuals (five female) who underwent clinical evaluation for drug-resistant epilepsy participated in the study. Electrodes were implanted intracranially (iEEG) with the clinical goal of identifying epileptogenic foci for surgical removal. Any electrodes showing signs of epileptiform discharges, as identified by an epileptologist, were not analysed in this study. Before electrode implantation, all participants provided written informed consent for participation in the study. The research protocol was approved by the Institutional Review Board of the North Shore University Hospital.

The participants listened to naturalistic recordings of voice actors reading passages from stories and conversations. To ensure that the participants were paying attention to the stimuli, one of the voices in the recording occasionally directed a question directly at the listener, or the stories were paused, and the participant was asked a question to check their understanding. The participants were able to answer each question effectively. These pauses separated the stimulus into separate passages.

The envelope of the high-gamma band (70–150 Hz) of the raw neural recordings was computed using the Hilbert transform[63] and downsampled to 100 Hz. This signal was used as the neural response due to its correlation with neuronal firing rates[64,65] and its common use in auditory neuroscience research[66,67]. We restricted our analysis to speech-responsive electrodes, which we estimated using a *t*-test between the response of each electrode to the first second of the stimulus compared to the last second of silence preceding it (N samples = 100; false discovery rate corrected; $P < 0.05$ (ref. 20)), leaving 707 electrodes for analysis. We extracted average word responses from each electrode by taking the average high-gamma signal value in a 100-ms window around the midpoint of each word.

Large language models

We analysed 12 LLMs with approximately seven billion parameters downloaded from Hugging Face and implemented with its Transformers library[68], including the most recent and most popular open-source LLMs. We selected these models by searching the Hugging Face Hub for 7 billion parameter models, and then using as many of the trending or most downloaded models that we were able to run without issue.

We computed two evaluation metrics similar to those used by LLaMA2 (ref. 19): Reading Comprehension and Commonsense Reasoning. As measures of English language understanding, these are both highly related to the listening comprehension task performed by the human participants in the study. As in a previous study[19], these metrics were created by averaging the performance of a model on a certain set of related tasks. All individual benchmarks were computed for each model using the Language Model Evaluation Harness[69] on GitHub.

- Reading Comprehension: this metric was the average 0-shot performance of a model on SQuAD 2.0 (ref. 70) and BoolQ[71].
- Commonsense Reasoning: this metric consists of the average 0-shot performance on OpenBookQA[72], PIQA[73], HellaSwag[74] and WinoGrande[75].

Overall LLM performance was computed as the average Reading Comprehension and Commonsense Reasoning scores.

The models used, their benchmark performance and the overall LLM performance scores are shown in Table 1.

To extract LLM embeddings for each stimulus passage (approximately 30–60 s when spoken), we fed the text to the model and extracted the embeddings of each layer when given a causal attention mask. When limiting the contextual window of the model, the attention mask was truncated to include only the most recent N tokens. For multi-token words, we used the embedding of the last token in the word. Thus, for each passage, we extracted a tensor of embeddings of shape (L_{layers}, N_{words} and $D_{\text{dimensions}}$) from each model, where L_{layers} was 32 for each model.

Ridge regression mapping from embeddings to neural responses

We performed PCA to reduce the dimensionality of the embeddings of each model to 500 components. For a given model, PCA was performed separately for each layer. PCA was performed prior to ridge regression to minimize data storage and computation requirements for such a large number of embeddings. We then fitted tenfold cross-validated ridge regression models to predict the average word responses from the embeddings of each layer, sweeping over a range of regularization parameters for each training fold, using Scikit-learn's RidgeCV model[76].

Electrode localization and brain plotting

The electrode positions of each participant were mapped to the participant's brain using iELVis[77] to perform co-registration between pre- and post-implant magnetic resonance imaging scans. The participant-specific electrode locations were then mapped to the FreeSurfer average brain[78]. The distance from the posteromedial HG (TE1.1)[24] was computed in this average brain, because TE1.1 is a landmark of the primary auditory cortex[25–27,46]. We used the geodesic distance along the cortical surface from the pmHG to the nearest surface point of each electrode, computed using the SurfDist package[79], because it provides a strong estimate of neural connectivity without over-parameterizing a model of the hierarchy and adding extra assumptions. All subdural electrodes were snapped to the nearest surface point when visualizing electrodes on the average brain.

Estimating hierarchical alignment with layer centre of mass

To estimate the alignment between the hierarchy of the brain and the LLM layers, we used a method that allowed us to identify the depth in the layers of the LLM that best corresponded to the neural responses of the brain in each hierarchical region. First, we binned electrodes by location along the hierarchy of the brain, either with distance from the pmHG or with electrode latency. We then took all of the LLM layer-wise similarity scores for a single model for those electrodes, producing a matrix of shape ($N_{\text{electrodes}}$, 32) because the LLMs all have 32 layers. Normalizing the scores of each electrode between 0 and 1 and averaging over the electrodes in the group produced a vector of length 32 containing the average brain similarity score of this group of electrodes at each layer of this LLM. This vector can be seen in one row of one of the matrices in Fig. 3a. Treating these 32 values as a distribution, we computed the centre of mass over the layers, which produced a larger value when the peak of the averaged brain similarity was deeper in the layers of the model. Thus, a larger centre of mass value for a given bin

of electrodes indicates that these electrodes are most similar to the later layers of the LLM, and thus most strongly correspond to the later stages of the hierarchy of the LLM. Therefore, this method defines the relationship between a group of electrodes at a given stage of auditory processing and the depth in the LLM at which they display the greatest similarity. Because we used the centre of mass of the averaged scores rather than the peak layer indices of single electrodes, the centre of mass values typically ranged between layers 12 and 20. This is because the average distribution over a group of electrodes was never skewed enough that it produced a centre of mass in the earliest or latest layers of the model. The alignment between the hierarchy of the brain and the layer-wise hierarchy of the LLM could then be estimated as the correlation between the distance of a group of electrodes from the pmHG and the LLM layer centre of mass of that group. It should be noted that these raw hierarchy alignment correlation values should not be interpreted except in relation to each other, because the binning method produces non-independent samples, inflating the correlation, and was used to overcome the issue of non-uniform sampling of electrodes across the hierarchy.

Comparing LLMs with CKA

To estimate the similarity between high-dimensional embeddings of different models, we used CKA[30], a similarity metric that is related to CCA but has been shown to perform well in high-dimensional scenarios between neural network features. We used the radial basis function kernel to allow for nonlinear similarity measurement. For a given pair of models, we computed the CKA similarity between the embeddings of one layer of the first model and another layer of the second model. Iterating all the pairs of layers for these two models produced a single similarity matrix. These similarity matrices were then grouped by whether they described a comparison between two models in the top 5 of all LLMs for benchmark performance, one model in the top 5 and the other in the bottom 5 or two models in the bottom 5, and then averaged.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Although the iEEG recordings used in this study cannot be made publicly available due to patient privacy restrictions, they can be requested from the author (N.M.). A single electrode of example data is available alongside the code demo via Code Ocean at https://doi.org/10.24433/CO.0003780.v1 (ref. 80). Source data are provided with this paper.

Code availability

The code for preprocessing neural recordings, including extracting the high-gamma envelope and identifying responsive electrodes, is available in the naplib-python package[81]. The code for computing brain similarity scores with ridge regression is available via Code Ocean at https://doi.org/10.24433/CO.0003780.v1 (ref. 80).

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Author contributions

G.M. and Y.A.L. contributed to the conceptualization, methodology, data analysis and writing of the manuscript. S.B. and A.D.M. contributed to data collection. N.M. supervised the project and contributed to the conceptualization, methodology and revision of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software used outside of Tucker-Davis Technologies neurophysiology hardware.
Data analysis	We used the HuggingFace Transformers library v4 to extract LLM features. Naplib-python v1.4 was used for electrode selection and other analyses.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Although the iEEG recordings used in this study cannot be made publicly available due to patient privacy concerns, they can be requested from the author [N.M.].

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	Eight human subjects, 3 male and 5 female, participated in the study.
Population characteristics	The studied population consists of people in the hospital for neurological monitoring due to epilepsy.
Recruitment	Participants were recruited after they indicated a desire to participate in research while in the hospital, by speaking with the research coordinator of the hospital. Prior to electrode implantation, all subjects provided written informed consent for research participation.
Ethics oversight	The research protocol was approved by the institutional review board at North Shore University Hospital.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Eight subjects were recruited. This was determined by how many subjects were undergoing clinical epilepsy monitoring and consented to participate in research.
Data exclusions	No subjects were excluded.
Replication	Due to the limited availability of intracranial neural data, no reproducibility attempts have been made.
Randomization	N/A since no grouping was performed.
Blinding	N/A since no grouping was performed.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging